Fall 2025 ECEN 629 Homework Set 3

Prob. 1 Suppose $f : \mathbf{R}^n \to \mathbf{R}$ is convex, and define

$$g(x,t) = f(x/t), \quad \mathbf{dom} g = \{(x,t) | x/t \in \mathbf{dom} f, t > 0\}.$$

Show that g is quasiconvex, and then write down the optimization procedure similarly to that on page 17 of the slides of Chapter 4, specialized to this problem.

Hint: consider the function $t \cdot f(x/t)$.

Solution: For any a, the sublevel set $\{(x,t)|f(x/t) \leq a\} = \{(x,t)|tf(x/t) \leq at\} = \{(x,t)|tf(x/t) - at \leq 0\}$. This is the 0-level set of the function

$$h(x,t) = tf(x/t) - at.$$

$h(x,t)$ is clearly convex since it is a sum of the perspective of the convex function f and an affine function, and thus we know this set is convex for any a.

Prob. 2 Some non-convex problems can also be easy to solve. Find the optimal solution for the following problems in closed form.

(a) For $x \in \mathbb{R}^n$

$$\begin{aligned} \text{minimize: } & ||x||_2 \\ \text{subject to: } & 10 \geq ||x||_1 \geq 1. \end{aligned}$$

(b) Let $x = (x_1, x_2) \in \mathbb{R}^2$

$$\begin{aligned} \text{minimize: } & x_1 + x_2 \\ \text{subject to: } & x_1 + 3x_2 = 5; \\ & (x_1 - 3)^2 + (x_2 - 4)^2 \geq 25. \end{aligned}$$

Solution:

(a) By the symmetry, we can restrict our attention to the positive quadrant, i.e., $x_i \geq 0$, $i = 1, 2, \ldots, n$. The solution in this case is $(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n})$, and the optimal value is $p^* = \frac{1}{\sqrt{n}}$.

(b) This is unbounded below, and thus the optimal value is $-\infty$.

Prob. 3 Huber's penalty function is defined as

$$\phi(u) = \begin{cases} u^2 & |u| \leq M \\ M(2|u| - M) & |u| > M \end{cases}.$$

(a) Is the function $\phi(u)$ convex in general?

(b) Show that the minimization problem

$$\text{minimize: } \sum_{i=1}^{m} \phi(a_i^T x - b_i)$$

is equivalent to the following problem

$$\begin{aligned} \text{minimize: } & \sum_{i=1}^{m} (u_i^2 + 2Mv_i) \\ \text{subject to: } & -u - v \preceq Ax - b \preceq u + v \\ & 0 \preceq u \preceq M\mathbf{1} \\ & v \succeq 0. \end{aligned}$$

(c) A general form of Huber's penalty function with parameters (α, β, γ) is

$$\phi(u) = \begin{cases} \alpha \cdot u^2 & |u| \leq M \\ \beta M(\gamma|u| - M) & |u| > M \end{cases}.$$

For any fixed value of $\alpha > 0$, find the valid assignment of (β, γ) in terms of α such that the penalty function is convex.

Solution:

(a) Yes, it is convex. We can show this by checking the first order condition. More precisely, the first order derivative of the function is

$$\phi'(u) = \begin{cases} 2u & |u| \leq M \\ 2M & u > M \\ -2M & u < -M \end{cases}.$$

We need to show that for any x, y, $f(y) \geq f(x) + \nabla f(x)^T(y - x)$. We can now consider several cases: if $x > M$ and $y > M$, then the condition becomes

$$M(2y - M) \geq M(2x - M) + 2M(y - x), \tag{1}$$

which is trivially true. If $x > M$ and $y < -M$, the condition is

$$M(-2y - M) \geq M(2x - M) + 2M(y - x), \tag{2}$$

which is also trivially true. If $x > M$ and $|y| \leq M$, the condition is

$$y^2 \geq M(2x - M) + 2M(y - x), \tag{3}$$

which is equivalent to

$$y^2 - 2MY + M^2 \geq 0, \tag{4}$$

which is also trivially true. Other cases when $x < -M$ and $|x| \leq M$ can also be checked easily.

(b) The first observation is that in the latter problem, for fixed x, at the optimum we must have $u_i + v_i = |a_i^T x - b_i|$, since otherwise we can decrease u_i or v_i without violating the constraints for a lower objective function value. Now substituting $v_i = |a_i^T x - b_i| - u_i$, we have the following equivalent optimization problem

$$\begin{aligned} \text{minimize: } & \sum_i^m (u_i^2 - 2Mu_i + 2M|a_i^T x - b_i|) \\ \text{subject to: } & 0 \le u_i \le \min\{M, |a_i^T - b_i|\}, \quad i = 1, \dots, m. \end{aligned}$$

This optimization problem is separable. Inspecting the constraint leads to the conclusion that it is equivalent to the original problem.

(c) Since $\phi(\mu)$ is convex, we must have the function to be continuous and its derivative to be monotonically non-decreasing. The first condition gives

$$\alpha = \beta(\gamma - 1).$$

The second condition gives

$$2\alpha \le \beta\gamma.$$

Prob. 4 Write the following problems as equivalent LPs:

(a)

$$\begin{aligned} \text{minimize: } & \|x\|_\infty \\ \text{subject to: } & ||Ax - b||_1 \le 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(b)

$$\begin{aligned} \text{minimize: } & \|x\|_\infty \\ \text{subject to: } & ||Ax - b||_\infty \le 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(c)

$$\begin{aligned} \text{minimize: } & \|x\|_1 \\ \text{subject to: } & ||Ax - b||_\infty \le 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(d)

$$\begin{aligned} \text{minimize: } & \|x\|_1 \\ \text{subject to: } & ||Ax - b||_1 \le 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

For the problems above, what if we exchange the objective function and the constraint, can you still write down the problems as LPs?

Solution:

(a) The problem is equivalent to the LP

$$\begin{aligned} \text{minimize: } & y \\ \text{subject to: } & -y \cdot \mathbf{1} \preceq x \preceq y \cdot \mathbf{1} \\ & -t \preceq Ax - b \preceq t \\ & t^T \mathbf{1} \leq 1 \end{aligned}$$

where **1** is an all 1 vector.

(b) The problem is equivalent to the LP

$$\begin{aligned} \text{minimize: } & y \\ \text{subject to: } & -y \cdot \mathbf{1} \preceq x \preceq y \cdot \mathbf{1} \\ & -\mathbf{1} \preceq Ax - b \preceq \mathbf{1}. \end{aligned}$$

where **1** is an all 1 vector.

(c) The problem is equivalent to the LP

$$\begin{aligned} \text{minimize: } & \mathbf{1}^T y \\ \text{subject to: } & -y \preceq x \preceq y \\ & -\mathbf{1} \preceq Ax - b \preceq \mathbf{1}. \end{aligned}$$

where **1** is an all 1 vector.

(d) The problem is equivalent to the LP

$$\begin{aligned} \text{minimize: } & \mathbf{1}^T y \\ \text{subject to: } & -y \preceq x \preceq y \\ & -t \preceq Ax - b \preceq t \\ & t^T \mathbf{1} \leq 1 \end{aligned}$$

Remark: pay attention to the newly introduced decision variables y, t, and recognize their dimensions (i.e., which is a scalar and which is a vector).

Prob. 5 Consider the optimization problem in $\mathbb{R}^2$

$$\begin{aligned} \text{minimize: } & f_0(x_1, x_2) \\ \text{subject to: } & (x_1 - 1)^2 + (x_2 - 2)^2 \leq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0,\ x_2 \geq 0. \end{aligned}$$

First make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value, using two methods: geometrically and the first-order optimality condition. Note that you should not use KKT directly here, but the first-order condition in Chapter 4.

(a) $f_0(x_1, x_2) = x_1 + x_2$;

(b) $f_0(x_1, x_2) = x_1^2 + x_2^2$.

Solution:

(a) The sketch is omitted. The intersection of the line $2x_1 + x_2 = 3$ and the circle $(x_1 - 1)^2 + (x_2 - 2)^2 = 2$ are $(0, 3)$ and $(\frac{6}{5}, \frac{3}{5})$. From the geometry, the optimal solution is clearly at the corner point $(\frac{6}{5}, \frac{3}{5})$, and thus $p^* = \frac{9}{5}$. Let us next check if the first order optimality condition is true: $\nabla f_0 = (1, 1)$. Therefore, we need to check if

$$(1, 1)(y_1 - 6/5, y_2 - 3/5)^T \geq 0 \tag{5}$$

for all feasible y. Next we prove this by contradiction. Suppose there is a feasible point such that

$$(1, 1)(y_1 - 6/5, y_2 - 3/5)^T < 0, \tag{6}$$

i.e., $y_1 + y_2 < 9/5$ which when combined with the second constraint, gives $y_1 > 6/5$. However, with the first constraint, this implies that $|y_2 - 2| < 7/5$, i.e., $3/5 < y_2 < 17/5$. This leads to the contradiction that $y_1 + y_2 > 9/5$.

The proof by contradiction can also be avoided if we partition the feasible set into different parts and verify the tangent direction of the boundaries.

(b) From the geometry, it is not hard to determine whether the optimal solution is the tangent point of the circle $x_1^2 + x_2^2 = r$ with the line $2x_1 + x_2 = 3$, or the corner point $(\frac{6}{5}, \frac{3}{5})$, and thus we need to find out. The tangent point can be calculated as the intersection of the line $-x_1 + 2x_2 = 0$, which is $(\frac{6}{5}, \frac{3}{5})$, i.e., they are actually the same point. Thus the optimal solution is again $(\frac{6}{5}, \frac{3}{5})$, and thus $p^* = \frac{9}{5}$.

We can again use the first order optimality condition to confirm this. Here we need to show that

$$(12/5, 6/5)(y_1 - 6/5, y_2 - 3/5)^T \geq 0, \tag{7}$$

but this is equivalent to

$$2y_1 + y_2 \geq 3, \tag{8}$$

which is clearly true due to the second constraint in the problem.

Prob. 6 Consider the optimization problem

$$\begin{aligned} \text{minimize: } & f_0(x_1, x_2) \\ \text{subject to: } & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0,\ x_2 \geq 0. \end{aligned}$$

First make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value, using two methods: geometrically and the first-order optimality condition.

(a) $f_0(x_1, x_2) = 3x_1 + 2x_2$;

(b) $f_0(x_1, x_2) = \max\{x_1, x_2\}$.

Solution: The region has three corner points at $(0, 3), (1, 1), (2, 0)$.

(a) $3x_1 + 2x_2 = b$ is a hyperplane, and this is a typical LP problem. The solution is $(1, 1)$, and $p^* = 5$. We can use the same argument as in the previous question to apply the first order optimality condition, which requires $3y_1 + xy_2 \geq 5$ for any feasible y. This is clearly true by summing the first two constraint together.

(b) The optimal point is $(1, 1)$, and $p^* = 1$. There are several ways to see this:

- One way is to view it as consisting of two optimization problems. One is to optimize x_1, subject to the original constraints and the additional constraint $x_1 \geq x_2$; another one is to optimize x_2, subject to the original constraints and $x_1 \leq x_2$. The minimum of the solution of the two problems is the solution of the original problem.
- Another way is to use the ℓ_∞ norm ball, since we know the feasible set has $x_1 \geq 1$ and $x_2 \geq 1$, and thus with this set of constraint, we can equivalent consider $f_0(x_1, x_2) = ||x||_\infty$. Now drawing the ℓ_∞ norm ball sitting at $(0, 0)$ and touching the feasible set will lead to the answer.

Strictly speaking, the objective function is not differentiable, thus the first order condition is not directly applicable to this problem. If you have used sub-differential (subgradient), then you can still derive sub-differential, and used a general version of the first-order optimality condition to verify the optimality. Alternatively, we can consider the following equivalent problem

$$\begin{aligned} \text{minimize: } & t \\ \text{subject to: } & x_1 \leq t,\ x_2 \leq t \\ & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0,\ x_2 \geq 0. \end{aligned}$$

Then the first order condition is

$$t - 1 \geq 0.$$

Let's again prove by contradiction: suppose there exist $t < 1$, then $x_1 < 1$ and $x_2 < 1$, however this is clearly not feasible with either $x_1 + x_2 \geq 2$ or $2x_1 + x_2 \geq 3$. Therefore the supposition cannot be true.

Prob. 7 Give explicit solutions for the following LP problems over $\mathbf{R}^n$

(a)

$$\begin{aligned} \text{minimize: } & c^T x \\ \text{subject to: } & a^T x \leq b, \end{aligned}$$

where $a \neq 0$.

(b)

$$\begin{aligned} \text{minimize: } & c^T x \\ \text{subject to: } & \mathbf{1}^T x \leq \alpha \\ & 0 \cdot \mathbf{1} \preceq l \preceq x \preceq u. \end{aligned}$$

Solution:

(a) The solution depends on whether a and c are essentially the same direction. If there does not exist a λ such that $\lambda a = c$, then the hyperplanes $c^T x = 0$ and $a^T x = b$ are not parallel, which means that the problem is unbounded. On the other hand, if there exists such a λ, then the two hyperplanes are parallel. In this case, we need to consider if λ is positive or negative: only when $\lambda \leq 0$, the problem is bounded and the optimal value is equal to λb; otherwise again the problem is unbounded.

(b) Without loss of generality, assume $c_1 \leq c_2 \leq \ldots \leq c_k \leq 0 \leq c_{k+1} \leq \ldots \leq c_n$, since otherwise we can re-order the indices to find an equivalent problem. We can define the variable $y_i = x_i - l_i$, and rewrite the problem as

$$\begin{aligned} \text{minimize: } & c^T y + c^T l \\ \text{subject to: } & \mathbf{1}^T y + \mathbf{1}^T l \leq \alpha \\ & 0 \cdot \mathbf{1} \preceq y \preceq u - l. \end{aligned}$$

If $\alpha < \mathbf{1}^T l$, then the problem is not feasible, and the optimal value is ∞. Otherwise, $y = 0 \cdot \mathbf{1}$ is clearly in the feasible set. Using $y = 0 \cdot \mathbf{1}$ as the starting point, we consider increasing the values of y_i's to reduce the objective function. Clearly, we wish to choose those y_i's corresponding to negative c_i as large as possible, and keep all other $y_i = 0$. Thus we need to find an $i_0 \leq k$ such that $\sum_{i=1}^{i_0}(u_i - l_i) < \alpha - \mathbf{1}^T l \leq \sum_{i=1}^{i_0+1} y_i$. Then the optimal point is to let

$$\begin{aligned} & y_i = u_i - l_i, \quad i = 1, 2, \ldots, i_0, \\ & y_{i_0+1} = \alpha - \mathbf{1}^T l - \sum_{i=1}^{i_0} y_i, \\ & y_i = 0, \quad i = i_0 + 2, \ldots, n. \end{aligned}$$

Prob. 8. Find the dual optimization problems for the following optimization problems

(a)

$$\begin{aligned} \text{minimize: } & c^T x \\ \text{subject to: } & Ax = b \\ & \ell \preceq x \preceq u, \end{aligned}$$

where $x \in \mathbf{R}^n$ is the variable, and $\ell, u \in \mathbf{R}^n$.

(b) Recall last time you wrote the following problem as an equivalent LP:

$$\begin{aligned} \text{minimize: } & \|x\|_\infty \\ \text{subject to: } & ||Ax - b||_\infty \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Find the dual problem for the equivalent LP.

(c)

$$\begin{aligned} \text{minimize: } & \sum_{i=1}^n x_i \log x_i \\ \text{subject to: } & Ax \succeq b \\ & c^T x = d, \end{aligned}$$

whose domain is R_+^n, i.e., $x_i > 0$ for all $i = 1, 2, \ldots, n$. Assume $c \succeq 0$.

(d)

$$\begin{aligned} \text{minimize: } & x^T Q x \\ \text{subject to: } & Ax = b, \end{aligned}$$

where Q is a positive definite matrix.

Solution:

(a) Dual derivation:
Introduce Lagrange multipliers λ for equality constraints, $\mu \succeq 0$ for $x \succeq \ell$, and $\nu \succeq 0$ for $x \preceq u$.

$$\begin{aligned} \mathcal{L}(x, \lambda, \mu, \nu) &= c^T x + \lambda^T (Ax - b) + \mu^T (\ell - x) + \nu^T (x - u) \\ &= (c + A^T \lambda - \mu + \nu)^T x - b^T \lambda + \mu^T \ell - \nu^T u. \end{aligned}$$

The dual function is finite iff $c + A^T \lambda - \mu + \nu = 0$. Then

$$g(\lambda, \mu, \nu) = -b^T \lambda + \mu^T \ell - \nu^T u.$$

Dual problem:

$$\begin{aligned} \max_{\lambda, \mu, \nu} \quad & -b^T \lambda + \mu^T \ell - \nu^T u \\ \text{s.t.} \quad & c + A^T \lambda - \mu + \nu = 0, \\ & \mu, \nu \succeq 0. \end{aligned}$$

(b) Primal problem:

$$\begin{aligned} \min_{x, y} \quad & y \\ \text{s.t.} \quad & -y\mathbf{1} \preceq x \preceq y\mathbf{1}, \\ & -\mathbf{1} \preceq Ax - b \preceq \mathbf{1}. \end{aligned}$$

Dual derivation:
Let $u, v \succeq 0$ correspond to $Ax - b \preceq \mathbf{1}$ and $-Ax + b \preceq \mathbf{1}$, and $p, q \succeq 0$ to $x \preceq y\mathbf{1}$ and $-x \preceq y\mathbf{1}$.
Lagrangian:

$$\begin{aligned} \mathcal{L} &= y + p^T (x - y\mathbf{1}) + q^T (-x - y\mathbf{1}) + u^T (Ax - b - \mathbf{1}) + v^T (-Ax + b - \mathbf{1}) \\ &= (p - q + A^T (u - v))^T x + (1 - \mathbf{1}^T (p + q)) y - (\mathbf{1}^T (u + v) + b^T (u - v)). \end{aligned}$$

Finite infimum conditions:

$$p - q + A^T (u - v) = 0, \quad \mathbf{1}^T (p + q) = 1.$$

Dual function:

$$g(u, v, p, q) = -\mathbf{1}^T (u + v) - b^T (u - v).$$

Dual problem:

$$\begin{aligned} \max_{u, v, p, q \succeq 0} \quad & -\mathbf{1}^T (u + v) - b^T (u - v) \\ \text{s.t.} \quad & A^T (u - v) + p - q = 0, \\ & \mathbf{1}^T (p + q) = 1. \end{aligned}$$

(c) Primal problem:

$$\begin{aligned} \min_{x>0} \quad & \sum_{i=1}^{n} x_i \log x_i \\ \text{s.t.} \quad & Ax \succeq b, \\ & c^T x = d. \end{aligned}$$

Dual derivation:

$$\begin{aligned} \mathcal{L}(x, \lambda, \nu) &= \sum_i x_i \log x_i + \lambda^T (b - Ax) + \nu (c^T x - d) \\ &= \sum_i [x_i \log x_i - x_i (A^T \lambda)_i + \nu c_i x_i] + \lambda^T b - \nu d. \end{aligned}$$

Optimal x_i satisfies $\log x_i + 1 - (A^T \lambda)_i + \nu c_i = 0$, hence $x_i = \exp((A^T \lambda)_i - \nu c_i - 1)$. Substituting back,

$$g(\lambda, \nu) = \lambda^T b - \nu d - \sum_i e^{(A^T \lambda)_i - \nu c_i - 1}.$$

Dual problem:

$$\max_{\lambda \succeq 0, \nu} \quad \lambda^T b - \nu d - \sum_{i=1}^{n} e^{(A^T \lambda)_i - \nu c_i - 1}.$$

(d) Primal problem:

$$\begin{aligned} \min_{x} \quad & x^T Q x \\ \text{s.t.} \quad & Ax = b. \end{aligned}$$

Dual derivation:

$$\mathcal{L}(x, \lambda) = x^T Q x + \lambda^T (Ax - b).$$

Stationarity gives $2Qx + A^T \lambda = 0$, or $x = -\frac{1}{2} Q^{-1} A^T \lambda$. Substitute back:

$$g(\lambda) = -\tfrac{1}{4} \lambda^T A Q^{-1} A^T \lambda - b^T \lambda.$$

Dual problem:

$$\max_{\lambda} \quad -\tfrac{1}{4} \lambda^T A Q^{-1} A^T \lambda - b^T \lambda.$$

Note here λ is the dual variable for the equality constraint, therefore, it is not required for it to be non-negative in the dual problem.